

# RAHUL RAO

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## EDUCATION

### Northeastern University

Boston, MA

Bachelor of Science in Computer Science, 3.9/4.0 GPA, Dean's List (All Semesters)

Expected Graduation: Dec. 2027

**Relevant Coursework:** Artificial Intelligence, Algorithms & Data Structures, Programming in C++, Discrete Structures, Matrix Methods for ML, Object-Oriented Design, Computer Systems, Multivariable Calculus, Linear Algebra, Principles of Mathematics, Theory of Computation

**Languages/Frameworks:** C++, Python, C, C#, Java, JavaScript, React, Node/Express, PyTorch, TensorFlow

**Developer Tools/Libraries:** Git, Google Colab, Docker, Apache Kafka, BigQuery, pandas, Matplotlib, SciPy

## EXPERIENCE

### Software Engineer Co-op - Order Experience (OX) Financials & Integration Team

Jul. 2026 – Dec. 2026

Wayfair

Boston, MA

- Replaced legacy **Kafka** pipelines with direct **BigQuery** writes in **C#**, **reducing ingestion-to-query latency** across **5M+ daily order events**.
- Built OX UI features to query **outstanding balance data**, letting **10+ teams** self-serve and removing recurring engineering requests.
- Built **reusable tax-calculation infrastructure** supporting **multi-entity fee logic** across backend services, launched with North Carolina's White Goods Disposal Fee and **designed to onboard future state fees** without added backend work.

### ML Research Assistant - Market Data Modeling

Jan. 2026 – Present

Northeastern University, Algorithms and Theory Research Group

Boston, MA

- Engineered a PyTorch temporal nowcasting model (**shared GRU + cross-sectional attention**) on 3 months of 1-second bar NIFTY data.
- Used **spot-futures lead-lag** structure to correct for stale prices, reaching a walk-forward  $R^2$  of **0.28** vs. **0.21** for a Ridge baseline.
- Constructed a **cross-validation training framework** to verify model scalability across noisy high-frequency environments.

### Data Science Foundations - Teaching Assistant

Jan. 2026 – May 2026

Northeastern University, Hourly College of Computer Science

Boston, MA

- Ran weekly office hours covering core ML concepts (kNN, linear regression) for **200+** sophomores and juniors.
- Graded weekly assignments and exams for **200+ students** with detailed Python-focused feedback.

### Software Engineering Intern

May 2024 – Aug. 2024

Hypatia AI

San Diego, CA

- Built **differential-equation simulations** in **Python** (SciPy/NumPy) to model insulin dynamics from glucose data.
- Integrated **Python backend functions** into iOS/Android apps to process CGM data in real time.

### Allayee Lab - Research Assistant

Apr. 2023 – May 2024

University of Southern California, Keck School of Medicine

Los Angeles, CA

- Built a pipeline to parse and normalize **30M+** rows of high-dimensional Genome-Wide Association Studies data for statistical analysis.
- Refactored Pheweb modules into an **interactive visualization tool** for USC researchers to explore **large-scale genetic associations**.

## ACTIVITIES

### Northeastern Blockchain

Jan. 2026 – Present

Director of Technology

Boston, MA

- Lead a **React** frontend rebuild and architecture update of the club's web platform with a team of 5 engineers.
- Designed curriculum and lead workshops for **40+ underclassman** on **smart-contract** development and **blockchain fundamentals**.

### Northeastern AI (AINU)

Sep. 2025 – Present

Director of Technology

Boston, MA

- Created a **technical ML track**, connecting a small cohort of students with professors for **hands-on academic research**.
- Lead a team of student engineers building AI projects and hosted hands-on **mathematics and ML workshops**.

## PROJECTS

### Legal Document Intelligence Pipeline | Python, PyTorch, Transformers, LangChain, Vector DBs

Sep. 2025 – Present

- Engineered a **Retrieval-Augmented Generation (RAG)** system over complex legal datasets, combining **semantic vector search** and **chunking strategies** to extract **structured domain insights**.
- **Tuned embedding and retrieval indexing** to improve answer relevance and cut **query latency by 85%** on large document sets.

### Optimum Hacknet - First Place Winner | RLNC, gRPC, P2P Networks

Sep. 2025

- Built a decentralized file-transfer system on Optimum's mump2p infrastructure, using **random linear network coding**.
- Integrated Go components behind a service proxy and helped **resolve network-latency issues** with Optimum's engineering team.

### AlgoTutorAI | Docker, Ollama API, REST API

Jun. 2025

- Used the **Ollama API** to build a context-aware tutoring engine that gives logic-based critiques of student solutions.
- Wrote a REST backend with automated data seeding, containerized with **Docker**.